Chapter 1: Foundations of Probability

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Opening & Motivation

Learning Objectives

- Understand sample space, events, and set operations
- Apply counting principles: addition and multiplication
- Compute permutations and combinations
- State the Probability Axioms and derive basic properties
- Apply the Addition Rule and simple Inclusion–Exclusion

Why Does AI Need Probability?

- Classifiers output *probabilities*, not certainties
- Example: $P(\text{spam} \mid \text{email}) = 0.87$
- Uncertainty is central to machine learning
- Probability theory provides the language to describe and quantify uncertainty

Warm-Up Question

- **Question:** If you flip a coin twice, what is the probability of getting *at least one head*?
- Take a guess based on intuition
- We will return to verify this answer later (see Slide 20)

Roadmap for Today

- Sample Space & Events
- Counting Principles (Permutations & Combinations)
- Probability Axioms
- Addition Rule & Inclusion–Exclusion

Sample Space & Events

Random Experiment and Outcome

- **Random experiment:** a process whose result cannot be predicted with certainty
- **Outcome:** a single result of the experiment
- Examples:
  - Rolling a die
  - Randomly sampling a data point from a dataset
  - Checking whether an email is spam

Sample Space

- **Definition:** the set of all possible outcomes, denoted Ω or S
- Example: rolling a die $\rightarrow S = \{1, 2, 3, 4, 5, 6\}$
- Example: checking an email $\rightarrow S = \{\text{spam}, \text{not spam}\}$
- Sample spaces can be discrete or continuous

Example: Listing a Sample Space

- Flip a coin twice:

$$S = \{HH, HT, TH, TT\}$$

- Roll a die twice: 36 outcomes total, e.g.

$$S = \{(1,1), (1,2), \ldots, (6,6)\}$$

Concept Check #1

- **Question:** What is the sample space for checking whether *three* emails are spam?
- Think for 30 seconds. . .

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- Think for 30 seconds. . .

Answer:

$$S = \{SSS, SSN, SNS, SNN, NSS, NSN, NNS, NNN\} \quad \text{(8 outcomes)}$$

Event

- An **event** is a subset of the sample space
- Example: rolling an even number $\rightarrow E = \{2, 4, 6\}$
- Events can be combined using set operations:
  - Union ($\cup$)
  - Intersection ($\cap$)
  - Complement (A^c)

Example: Defining Events

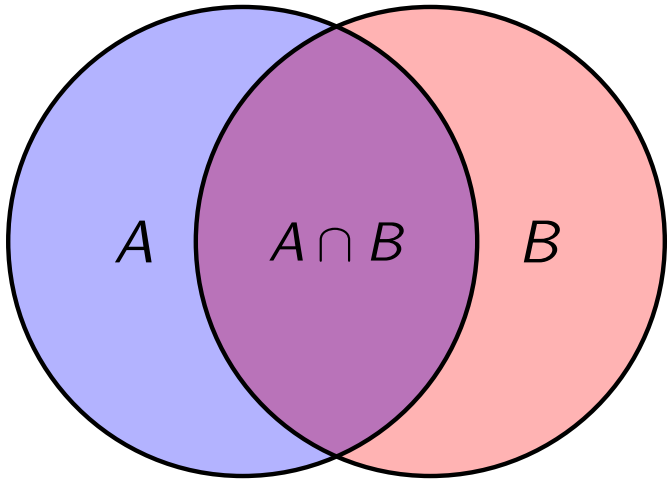
Recall $S = \{HH, HT, TH, TT\}$ (two coin flips).

- Event $A =$ "both flips are the same" $= \{HH, TT\}$
- Event $B =$ "at least one head" $= \{HH, HT, TH\}$

Set Operations: Union, Intersection, Complement

- $A \cup B$: outcomes in A *or* B (or both)
- $A \cap B$: outcomes in *both* A and B
- A^c: outcomes *not* in A

Venn Diagram Visualization



- Blue region: outcomes in A only $(A \cap B^c)$
- Red region: outcomes in B only $(A^c \cap B)$
- Purple region: outcomes in *both* $(A \cap B)$
- All three colored regions together $= A \cup B$

Example: Computing $A \cup B$ and $A \cap B$

Using $A = \{HH, TT\}$ and $B = \{HH, HT, TH\}$:

- $A \cup B = \{HH, HT, TH, TT\}$
- $A \cap B = \{HH\}$

Practice: Write out A^c and B^c.

Events in an AI Context

- A: model predicts positive class
- B: true label is positive
- $A \cap B$: **True Positive**
- $A \cap B^c$: **False Positive**
- This sets up Precision/Recall and Bayes' Theorem (Week 2)

Events in an AI Context

- A: test/model predicts **positive**
- B: the **true** condition is positive
- Four possible outcomes, defined via set operations:
  - $A \cap B$: **True Positive (TP)** — predicted positive, actually positive
  - $A \cap B^c$: **False Positive (FP)** — predicted positive, actually negative
  - $A^c \cap B$: **False Negative (FN)** — predicted negative, actually positive
  - $A^c \cap B^c$: **True Negative (TN)** — predicted negative, actually negative
- This sets up Precision/Recall (next slide) and Bayes' Theorem (Week 2)

Worked Example: COVID-19 Testing

Suppose 1000 people are tested for COVID-19. Let:

- B = "person actually has COVID" (**true condition**)
- A = "test result is positive" (**predicted condition**)

	Test Positive (A)	**Test Negative** (A^c)	**Total**
Actually Positive (B)	45 (TP)	5 (FN)	50
Actually Negative (B^c)	30 (FP)	920 (TN)	950
Total	75	925	1000

From the COVID Example: Precision and Recall

Using the table from the previous slide:

Precision — "Of those the test flagged positive, how many truly have COVID?"

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{45}{45 + 30} = \frac{45}{75} = 0.60$$

Recall (Sensitivity) — "Of those who truly have COVID, how many did the test catch?"

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{45}{45 + 5} = \frac{45}{50} = 0.90$$

Why This Matters

High recall (0.90) means the test catches most true cases — important for not missing infections. Lower precision (0.60) means many positive results are false alarms — something public health officials must also weigh (e.g., unnecessary quarantines).

Counting Principles

Addition Principle

- If task 1 can be done in m ways and task 2 in n ways,
- and the two tasks *cannot* occur together,
- then the total number of ways $= m + n$

Example: Addition Principle

- You choose *one* book: either from 5 AI books or 3 statistics books
- Total choices $= 5 + 3 = 8$

Multiplication Principle

- If a task consists of k steps, with $n_1, n_2, \ldots, n_k$ ways for each step,
- then the total number of ways:

$$n_1 \times n_2 \times \cdots \times n_k$$

Example: Multiplication Principle

- 3 binary features $\rightarrow 2^3 = 8$ possible combinations
- A 4-digit password (each digit 0–9) $\rightarrow 10^4$ possible codes

Revisiting the Warm-Up Question

Recall Slide 4: two coin flips.

- By the multiplication principle: $2 \times 2 = 4$ outcomes
- "At least one head" corresponds to 3 out of 4 outcomes

$\Rightarrow$ $P(\text{at least one head}) = \frac{3}{4}$

Permutations

- Number of ways to arrange r objects chosen from n distinct objects, **order matters**
- Formula:

$$P(n, r) = \frac{n!}{(n-r)!}$$

Example: Permutation

- Choose and order 3 samples out of 10 data points

$$P(10, 3) = \frac{10!}{7!} = 720$$

Combinations

- Number of ways to choose r objects from n distinct objects, **order does not matter**
- Formula:

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Example: Combination

- Choose 3 features out of 10 as model inputs

$$C(10,3) = \binom{10}{3} = 120$$

Permutation vs. Combination

	Order Matters?	**Formula**
Permutation	Yes	$\frac{n!}{(n-r)!}$
Combination	No	$\frac{n!}{r!(n-r)!}$

Comparative Example: Same Scenario, Two Methods

Choose 3 models out of 8:

- **No distinct roles** (ensemble):

$$C(8,3) = 56$$

- **Assign roles** (primary, secondary, backup):

$$P(8,3) = 336$$

- Whether order matters determines which formula to use

Practice: Permutation or Combination?

For each scenario, decide whether to use permutation or combination (no need to calculate):

1. Selecting the top 3 winners in a competition
2. Randomly selecting 3 people to form a team
3. Assigning 3 different tasks to 3 people
4. Choosing 3 flavors out of 10 for a sampler box

Probability Axioms

The Kolmogorov Axioms

For any event A in sample space S:

1. $P(A) \geq 0$ for every event A
2. $P(S) = 1$
3. If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B)$$

Example: Verifying the Axioms

- Rolling a fair die: $P(\{i\}) = \frac{1}{6}$ for $i = 1, \ldots, 6$
- Check Axiom 2:

$$P(S) = \sum_{i=1}^{6} \frac{1}{6} = 1 \quad \checkmark$$

Properties Derived from the Axioms

- $P(\emptyset) = 0$
- Complement rule: $P(A^c) = 1 - P(A)$
- Monotonicity: if $A \subseteq B$, then $P(A) \leq P(B)$
- $0 \leq P(A) \leq 1$

Example: Complement Rule

- Given $P(\text{spam}) = 0.3$
- Then

$$P(\text{not spam}) = 1 - 0.3 = 0.7$$

Example: Monotonicity

- $A =$ "rolling an even number" $= \{2, 4, 6\}$
- $B =$ "rolling a number greater than 1" $= \{2, 3, 4, 5, 6\}$
- Since $A \subseteq B$:

$$P(A) = \frac{3}{6} \leq P(B) = \frac{5}{6} \quad \checkmark$$

Concept Check #2

- If $P(A) = 0.6$, what is $P(A^c)$?
- If $A \subseteq B$ and $P(B) = 0.4$, what is the possible range of $P(A)$?

Concept Check #2

- If $P(A) = 0.6$, what is $P(A^c)$?
- If $A \subseteq B$ and $P(B) = 0.4$, what is the possible range of $P(A)$?

Answers: $P(A^c) = 0.4$; $\quad 0 \leq P(A) \leq 0.4$

Summary: Relationship Between Axioms and Properties

- The three axioms are the foundation
- All other properties (complement rule, monotonicity, bounds) are *derived*
- These properties will be used repeatedly throughout the course

Addition Rule & Inclusion–Exclusion

The Addition Rule (General Case)

For any two events A, B:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- If A and B are mutually exclusive, $P(A \cap B) = 0$
- The formula reduces to Axiom 3

Example: Addition Rule

In a class:

- 60% of students know Python
- 40% know R
- 25% know both

$$P(\text{Python} \cup R) = 0.6 + 0.4 - 0.25 = 0.75$$

Comprehensive Example: AI Classification Scenario

Given 100 data points:

- 40 are actually positive (A)
- 35 are predicted positive (B)
- 30 are both actually positive *and* predicted positive

Compute:

$$P(A) = 0.40, \quad P(B) = 0.35, \quad P(A \cap B) = 0.30$$

$$P(A \cup B) = 0.40 + 0.35 - 0.30 = 0.45$$

Question: What is $P(A^c \cap B)$ (False Positive)?

Simple Inclusion–Exclusion (Three Events)

For three events A, B, C:

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C) \end{aligned}$$

- You are only required to substitute values, not derive the formula
- More complex cases (4+ events) are beyond this course's scope

In-Class Practice (with Solutions)

Solve the following, then compare with the provided solutions:

1. Compute $C(12, 4)$
2. Compute $P(9, 2)$
3. Given $P(A) = 0.5, P(B) = 0.3, P(A \cap B) = 0.1$, find $P(A \cup B)$
4. Given $P(A) = 0.7$, find $P(A^c)$

Practice

C1: Pipeline Configurations with a Restriction

A model pipeline requires choosing:

- one preprocessing method (3 options: A, B, C)
- one feature selection method (4 options)
- one classifier (5 options)

However, 2 of the 5 classifiers are incompatible with preprocessing method A.

Question

How many valid pipeline configurations are there in total?

C2: Addition vs. Multiplication

A survey allows respondents to:

(a) pick *either* one programming language (5 choices) *or* one framework (4 choices), but not both, **or**

(b) pick *one language AND one framework*

Question

Compute the number of response options for (a) and for (b). Which counting principle applies in each case, and why?

C3: Ensemble Selection with a Conflict

You are selecting a 5-model ensemble from 12 candidate models. Models A and B are known to conflict and **cannot both** be selected.

Question

How many valid ensembles are possible?

- *Hint:* Consider computing the total number of ensembles, then subtracting the invalid ones.

C4: Validation Set with a Minimum Requirement

A validation set of 4 images must be chosen from 10 total images. It must include **at least 2** images from a "hard examples" subset of 4 images.

Question

How many valid validation sets are possible?

- *Hint:* Split into cases based on exactly how many hard examples are included.

C5: Leaderboard Arrangement with a Restriction

Six different models are ranked on a leaderboard (1st through 6th place). Two specific models, X and Y, are known in advance to be the strongest, but for fairness, the ranking rules require that X and Y **must not be adjacent** in the ranking.

Question

How many valid rankings are possible?

- *Hint:* Total arrangements minus arrangements where X and Y *are* adjacent.

C6 (Optional/Advanced): Prove the Addition Rule

Question

Using only the three probability axioms, prove that for any two events A, B:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- *Hint:* Partition $A \cup B$ into three mutually exclusive pieces: $A \cap B^c$, $A \cap B$, and $A^c \cap B$.

C7 (Optional/Advanced): Prove a Monotonicity Corollary

Question

Prove that if $A \subseteq B$, then

$$P(B \setminus A) = P(B) - P(A)$$

where $B \setminus A$ denotes outcomes in B but not in A.

C8: Beyond the Union — Exactly-One Region

A survey of 200 ML practitioners found:

- 120 use Python, 90 use R, 70 use Julia
- 50 use Python & R, 40 use Python & Julia, 30 use R & Julia
- 15 use all three

Questions

(a) How many use **at least one** language?
(b) How many use **exactly one** language?

- *Hint for (b):* You must subtract off the double- and triple-counted regions carefully — this goes beyond the standard union formula.

C9: Spot the Error — Counting

A classmate solves the following problem:

> *"How many ways can you choose a top-3 leaderboard (1st, 2nd, 3rd place) from 8 models?"*

Their answer: $C(8, 3) = 56$.

Question

Identify the error in their reasoning, and give the correct answer.

C10: Spot the Error — Probability

A classmate claims:

"Since $P(A) = 0.6$ *and* $P(B) = 0.5$*, then* $P(A \cup B) = 1.1$*."*

Question

Explain what is wrong with this reasoning, and state what additional information would be needed to compute $P(A \cup B)$ correctly.

Capstone Challenge: The Birthday-Month Problem

In a class of 30 students, assume each student's birth month is equally likely to be any of the 12 months (ignore leap years, twins, etc.).

Question

What is the probability that **at least two** students share the same birth month?

- *Hint:* Use the complement rule. Compute $P(\text{no shared month})$ first using the multiplication principle.
- Think carefully about what happens when the number of students exceeds the number of months...

Capstone Teaser: A Preview of Week 2

A spam filter's performance is described by:

$$P(\text{true spam}) = 0.10, \quad P(\text{flagged}) = 0.12, \quad P(\text{true spam} \cap \text{flagged}) = 0.09$$

Question

What fraction of *flagged* emails are actually true spam?

Summary

Summary

This section covers:

- Sample space, events, and set operations
- Counting principles: permutations & combinations
- Probability axioms and derived properties
- Addition Rule and Inclusion–Exclusion

Answer Key (Appendix)

The following slides contain full solutions to the Challenge Practice Set (C1–C10) and the Capstone problems.

Use these to self-check after attempting the problems independently.

Solution C1: Pipeline Configurations with a Restriction

Setup: 3 preprocessing $\times$ 4 feature selection $\times$ 5 classifiers, minus invalid combos.

$$\text{Total (unrestricted)} = 3 \times 4 \times 5 = 60$$

$$\text{Invalid (preprocessing A} + \text{2 bad classifiers)} = 1 \times 4 \times 2 = 8$$

$$\text{Valid configurations} = 60 - 8 = \boxed{52}$$

Solution C2: Addition vs. Multiplication

(a) "Either/or, not both" $\rightarrow$ **Addition Principle**:

$$5 + 4 = 9 \text{ options}$$

(b) "One AND the other" $\rightarrow$ **Multiplication Principle**:

$$5 \times 4 = 20 \text{ options}$$

Key distinction: Use addition when choices are mutually exclusive alternatives; use multiplication when choices are made independently in sequence/combination.

Solution C3: Ensemble Selection with a Conflict

Total ensembles (no restriction):

$$C(12, 5) = 792$$

Invalid ensembles (both A and B included): fix A and B, choose remaining 3 from the other 10 models:

$$C(10, 3) = 120$$

Valid ensembles:

$$792 - 120 = \boxed{672}$$

Solution C4: Validation Set with a Minimum Requirement

Split by number of hard examples included (2, 3, or 4):

$$\text{Exactly 2 hard} : C(4,2) \times C(6,2) = 6 \times 15 = 90$$
$$\text{Exactly 3 hard} : C(4,3) \times C(6,1) = 4 \times 6 = 24$$
$$\text{Exactly 4 hard} : C(4,4) \times C(6,0) = 1 \times 1 = 1$$

Total:

$$90 + 24 + 1 = \boxed{115}$$

Solution C5: Leaderboard Arrangement with a Restriction

Total arrangements of 6 models:

$$6! = 720$$

Arrangements where X and Y are adjacent: treat X, Y as a block (2 internal orders) with 5 remaining units:

$$2 \times 5! = 2 \times 120 = 240$$

Valid arrangements (not adjacent):

$$720 - 240 = \boxed{480}$$

Solution C6: Proof of the Addition Rule

Partition $A \cup B$ into three mutually exclusive sets:

$$A \cup B = (A \cap B^c) \cup (A \cap B) \cup (A^c \cap B)$$

By Axiom 3 (additivity for mutually exclusive events):

$$P(A \cup B) = P(A \cap B^c) + P(A \cap B) + P(A^c \cap B)$$

Since $A = (A \cap B^c) \cup (A \cap B)$ and $B = (A \cap B) \cup (A^c \cap B)$ (each a disjoint union):

$$P(A) = P(A \cap B^c) + P(A \cap B), \qquad P(B) = P(A \cap B) + P(A^c \cap B)$$

Adding and rearranging:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad \blacksquare$$

Solution C7: Monotonicity Corollary

Since $A \subseteq B$, we can write B as a disjoint union:

$$B = A \cup (B \setminus A), \qquad A \cap (B \setminus A) = \emptyset$$

By Axiom 3:

$$P(B) = P(A) + P(B \setminus A)$$

Rearranging:

$$P(B \setminus A) = P(B) - P(A) \qquad \blacksquare$$

(Note: this also proves monotonicity, since $P(B \setminus A) \geq 0$ by Axiom 1, hence $P(B) \geq P(A)$.)

Solution C8(a): At Least One Language

Using inclusion–exclusion for three sets:

$$P(\text{Py} \cup R \cup J) = 120 + 90 + 70 - 50 - 40 - 30 + 15$$
$$= \boxed{175 \text{ practitioners}}$$

Solution C8(b): Exactly One Language

Strategy: subtract all multi-language users from each single count.
Python only:

$$120 - 50 - 40 + 15 = 45$$

R only:

$$90 - 50 - 30 + 15 = 25$$

Julia only:

$$70 - 40 - 30 + 15 = 15$$

Total using exactly one language:

$$45 + 25 + 15 = \boxed{85 \text{ practitioners}}$$

Check: Exactly-one (85) + exactly-two + exactly-three should equal 175 from part (a). This is a good self-check exercise.

Solution C9: Spot the Error — Counting

The error: the classmate used $C(8,3)$, which ignores **order**. But 1st, 2nd, and 3rd place are **distinct positions** order matters.
Correct approach: use permutations.

$$P(8,3) = \frac{8!}{5!} = 8 \times 7 \times 6 = \boxed{336}$$

Solution C10: Spot the Error — Probability

The error: the classmate applied

$$P(A \cup B) = P(A) + P(B)$$

which is only valid when A and B are **mutually exclusive**. In general:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Missing information: $P(A \cap B)$ is needed. Without it, $P(A \cup B)$ cannot be determined exactly — only bounded:

$$\max(P(A), P(B)) \leq P(A \cup B) \leq \min(1, P(A) + P(B))$$

Here: $0.6 \leq P(A \cup B) \leq 1$.

Solution: Birthday-Month Problem

Key insight: there are only 12 months but 30 students.
By the **Pigeonhole Principle**, it is *impossible* for all 30 students to have distinct birth months at least one pair must share a month.
Formally: $P(12, 30)$ would require selecting 30 distinct items from only 12, which is impossible, so

$$P(12, 30) = 0 \quad \Rightarrow \quad P(\text{no shared month}) = \frac{P(12, 30)}{12^{30}} = 0$$

$$P(\text{at least one shared month}) = 1 - 0 = \boxed{1}$$

Takeaway: This is a certain event, not just highly likely a nice bridge between counting and probability.

Solution: Capstone Teaser (Preview of Week 2)

Using this week's tools, we can only describe the pieces:

$$P(\text{true spam}) = 0.10, \;\; P(\text{flagged}) = 0.12, \;\; P(\text{true spam} \cap \text{flagged}) = 0.09$$

The question asks for the proportion of flagged emails that are truly spam this is a **conditional probability**:

$$P(\text{true spam} \mid \text{flagged}) = \frac{P(\text{true spam} \cap \text{flagged})}{P(\text{flagged})} = \frac{0.09}{0.12} = 0.75$$

We haven't formally defined conditional probability yet this is exactly what Week 2 will cover, including **Bayes' Theorem**.